

iCloudEMS CV Engineer - Live Task Submission

1. How would you scale this from one live camera to 500 cameras streaming at once? Where would it break first?

To scale to 500 cameras, the monolithic script must be decoupled into a distributed microservices architecture using message brokers (e.g., Kafka) and scalable inference servers (e.g., NVIDIA Triton). The system would break first at the hardware inference bottleneck; running 500 parallel YOLO-Pose streams on a single node will exhaust VRAM and compute immediately. Resolving this requires lowering the processing frame rate (e.g., analyzing 3-5 FPS instead of 30), batching frames from multiple streams, and horizontally scaling across a GPU cluster.

2. How would you avoid double-counting or losing track of a person if they briefly leave the camera's view?

First, the tracking algorithm's memory buffer (e.g., `track_buffer` in BoT-SORT) must be extended to keep the track identity "alive" for several seconds during occlusions. For longer disappearances where the tracker drops the ID, a lightweight Re-Identification (Re-ID) mechanism is required. By extracting and storing an appearance embedding (like an HSV color histogram or a fast deep embedding) when a track exits, we can compare newly appearing tracks against recent exits and merge them if the similarity is high, preventing double-counting.

3. How would you handle a camera feed that's consistently blurry or poor quality - flag it, skip it, or something else?

Consistently poor feeds should not be blindly processed or silently skipped, as both pollute the database and hide hardware failures. The pipeline should calculate a rolling baseline of frame sharpness (using Laplacian variance) and automatically trigger a "Degraded Camera" alert to the IT/maintenance team if the median drops below an acceptable threshold. During this degraded state, the system should continue processing but explicitly tag all generated analytics in the database with a "Low Confidence" flag so downstream consumers know the data is unreliable until the lens is cleaned.